

Data Binding

Introduction to Data Binding

home.component.ts

```
1 export class HomeComponent {  
2   userName = "John";  
3   monthlyAmount = 5000;  
4  
5   calculateReturns() {  
6     const total = this.monthlyAmount * 12;  
7     return total;  
8   }  
9 }
```

Data

Logic

Methods

home.component.html

```
1 <h2>Welcome {{ userName }}</h2>  
2 <p>Monthly Amount: {{ monthlyAmount }}</p>  
3 <button>Calculate</button>
```

Data binding
is the nervous system



- 1- Interpolation `{{ }}` : display component data to the template បង្ហាញតម្លៃពី TS ទៅ html
- 2- Property binding `[]` : sets element properties based on component data. យកតម្លៃដាក់អោយ variable in the ts
- 3- Even binding `()` : responds to user actions and calls component methods សម្រាប់ហៅ method ប្រើអាស្រ័យតាមព្រឹត្តិការណ៍ដែលកើតឡើង
- 4- Two-way-binding : which combines property and event binding for form inputs. សម្រាប់ដាក់តម្លៃបង្ហាញរួមគ្នារវាង component and template

1 Interpolation

```
<h2>Welcome {{ userName }}</h2>;
```

2 Property binding

```
<button [disabled]="monthlyAmount <= 0">Calculate</button>
```

3 Event binding

```
<button (click)="calculateReturns()">Calculate</button>
```

4 Two-way binding

```
<input [(ngModel)]="monthlyAmount" />
```

Each type serves a different purpose and flows data in different directions

1 Interpolation

```
<p>Welcome, {{ userName }}</p>
<p>Today is: {{ currentDate }}</p>
```

2 Property binding

```
<button [disabled]="isInvalid">Submit</button>
<img [src]="profileImageUrl" />
```

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3 Event binding

```
<button (click)="saveForm()">Save</button>
<input (input)="onTyping($event)" />
<div (mouseenter)="highlight()">Hover me</div>
```

4 Two-way binding

```
<input [(ngModel)]="monthlyAmount" />
```

Why do we need Data binding ?



Data Binding

TypeScript file

```
1 export class HomeComponent {
2   userName = "John";
3   monthlyAmount = 5000;
4 }
```



HTML file

```
1 <h2>Welcome, {{ userName }}</h2>
2
3 <button [disabled]="monthlyAmount <= 0">
4   Calculate
5 </button>
```

When users interact with the view,
your data gets updated automatically



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This is the magic

Example on the binding :

1- Interpolation

- We can use the condition in the interpolation:

<p>Total months of investment: {{investmentPeriod * 12}}</p>

<p>Investment duration: {{investmentPeriod > 5? 'Long term' : 'Short term'}}</p>

- Call the method in the interpolation

TS:

//Simple calculation method

```
calculateYearlyInvestment(): number
{
    return this.monthlyAmount * 12;
}
```

} យើងធ្វើចំនួនតែមួយតែ simple calculation បើវាជា complex calculation it will effectively to the performance.

HTML:

<p>Yearly investment amount: \${calculateYearlyInvestment()}</p>

ពេលហៅវាមកប្រើ វានឹងបង្ហាញតម្លៃរបស់ដែល Return

Interpolation with Expressions

Interpolation is restricted — some expressions are **not allowed** for safety

- ✗ You cannot use **Assignments**
{{ monthlyAmount = 10000 }}
- ✗ You cannot use **Increment** or **Decrement operators**
{{ monthlyAmount++ }}
{{ investmentPeriod-- }}
- ✗ You cannot Create **new objects** with the new keyword
{{ new Date() }}
- ✗ You cannot access **global objects** like window or document
{{ window.location }}
{{ document.title }}

These **restriction** ensure that

- Predictable
- Secure

focusing on displaying data rather than modifying application state

Interpolation with Expressions

```
{{ 5000 * 12 }}
```

```
{{ investmentPeriod > 5 ? 'Long-term' : 'Short-term' }}
```

```
{{ calculateYearlyInvestment() }}
```

```
{{ "SIP Amount: " + monthlyAmount }}
```

while understanding the **limitations** and **best practices**

To protect data safety we use ?

TS:

```
//For safe navigation demo
```

```
fundName: string | null = null;
```

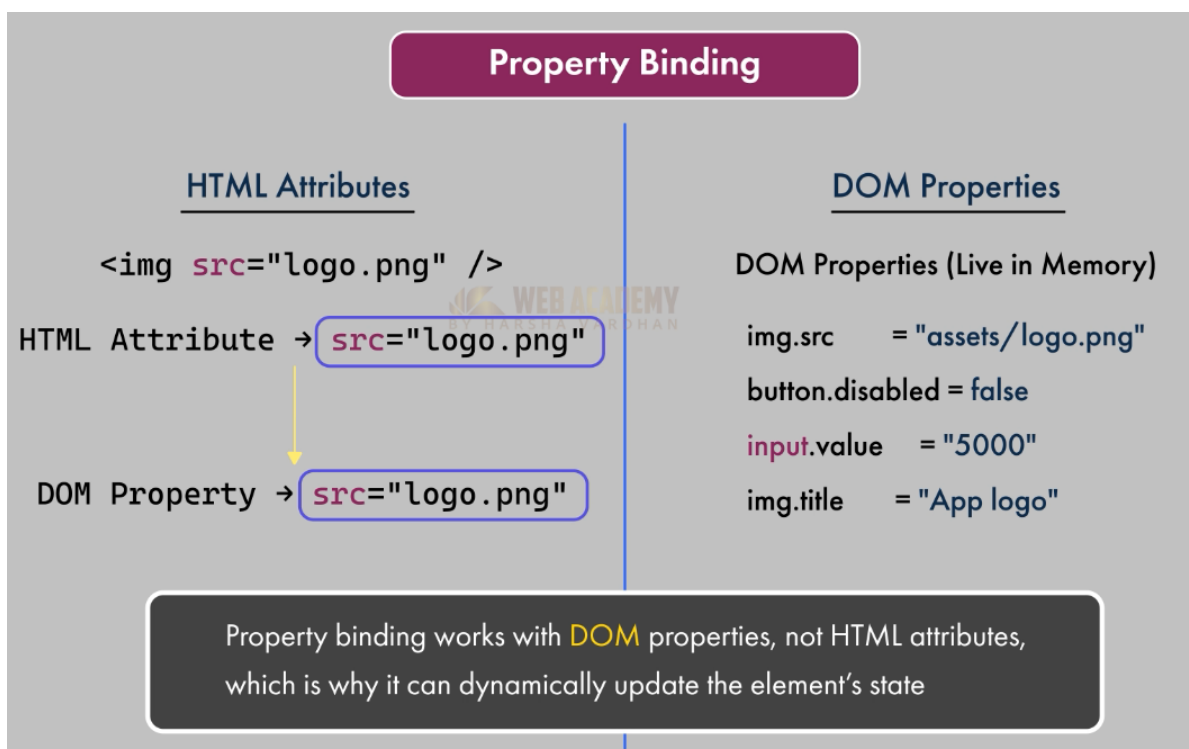
HTML:

```
<p>Fund name length: {{fundName?.length}}</p>
```

```
<th>{{investmentPeriod + ' years = ' + (investmentPeriod * 12) + ' months'}}</th>
```

⇒ 10 years = 120 months

2- Property Binding: when you need to control the HTML elements properties



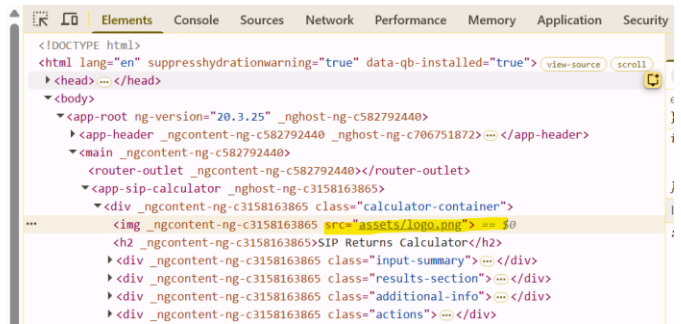
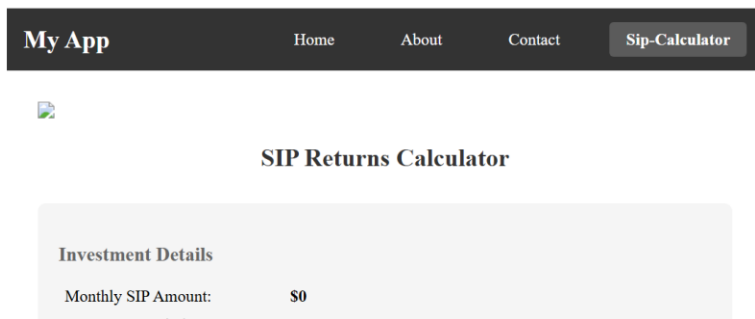
- Src in the html: ពេលវាដំណើរការឡើងមក src វាអោយតម្លៃទៅ Doc តែម្តង ហើយវារក្សាទុកនៅក្នុងនឹង វាមិនអាចធ្វើការផ្លាស់ប្តូរបានទេ
- [src] in the template: វានៅក្នុង Memory ចឹងពេលដែលយើងធ្វើការផ្លាស់ប្តូរ Value វា នឹងធ្វើការកែប្រែក្នុង Ram នឹង

```
<div class="actions">
```

```
<button [disabled]="monthlyAmount <=0 ">Calculate Return</button>
```

```
</div>
```

- monthlyAmount = 5000 so it is true : mean it is disable the button



- យើងប្រើប្រាស់ scr នឹងធ្វើការ assign តម្លៃអោយវា

- `<img [src]="logoUrl" />`

- logoUrl:string = "assets/logo.png"; នៅក្នុង component

- បង្កើតពេល mouse over ហើយអោយវាចេញ information [title] = component variable

`<div class="actions">`

`<button [disabled]="monthlyAmount <= 0" [title]="buttonToolTip">Calculate Return</button>`

`</div>`

Property Binding

One-way

Component (TS)

`monthlyAmount: number = 5000;`

↓

Template (HTML)

`[disabled]="monthlyAmount <= 0"`

Boolean Properties:

`true` → property set

`false` → property removed

- ចាំថា ពេលប្រើប្រាស់ Property Binding flow it from component to template (TS -> HTML)

`<div>`

`<input type="checkbox" [checked]="expectedReturnRate > 10"`

`id="highReturnCheckbox" />`

`<label for="highReturnCheckbox">High return investment</label>`

`</div>`

☒ High return investment

- id="highReturnCheckbox" and for="highReturnCheckbox" ពេលយើង click on the check box សម្រាប់ធ្វើការ ការចុត នឹងនៅលើ box ក៏ដើរ នៅលើ អក្សរវាង ដើរដល់

Property Binding

Angular Components

```
1 <app-parent>
2   <app-child [someCustomInput]="value"></app-child>
3 </app-parent>
```

We'll explore that when we learn about Component communication

Beginners mistake → Trying to use Interpolation where Property Binding is needed

Interpolation (Wrong ❌)

Property Binding (Right ✅)

```
<button disabled="{{ false }}">Calculate</button>
```

```
<button [disabled]="false">Calculate</button>
```

In HTML: attribute present = disabled, value doesn't matter



```
<button [disabled]="monthlyAmount <= 0">Calculate</button>
```

- យើងប្រើខុសដោយសារតែ ពេលយើងចង់ប្រើប្រាស់ Property binding តែយើងទៅយកតម្លៃមកពី Interpolation មកដាក់ចូល

3- Event – Binding (flow from template to component)

Event Binding - Click Events

Another thing to remember

Event Binding = Template → Component

HTML file

```
<button (click)="calculateReturns()">Calculate</button>
```

TypeScript file

```
calculateReturns() {
  console.log("Calculate button clicked!");
}
```

This separation of concerns makes your application **predictable** and **easier to debug**

Property Binding = Component → Template

TypeScript file

```
imagePath: string = "sample.png";
```

HTML file

```
<img [src]="imagePath" />
```

- `<button (click)="showEventDetails($event)">Show Event Details</button>`

```
// Method to show event details
showEventDetails(event: MouseEvent): void{
  console.log("Clicked at coordinates: ", event.clientX, event.clientY);
}
```

